

KO LANTA, Thailand

WMO: 485660

Lat: 7.533N

Lon: 99.050E

Elev: 3

StdP: 101.29

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	23.5	24.0	19.3	14.1	28.7	20.1	14.8	28.4	6.5	28.6	5.8	28.5	0.9	90	0.381

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	6.4	34.0	26.6	33.3	26.7	32.8	26.7	28.9	31.8	28.5	31.4	28.2	31.1	2.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.1	24.3	31.0	27.7	23.7	30.7	27.4	23.2	30.3	94.2	31.9	92.3	31.4	90.6	31.2	30.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	8.3	7.3	DB	22.1	35.1	0.9	0.9	21.5	35.8	21.0	36.3	20.5	36.8	19.9	37.5	(4)
(5)			WB	20.8	29.1	0.8	0.9	20.3	29.8	19.8	30.3	19.4	30.8	18.8	31.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	28.4	28.1	28.7	29.1	29.2	28.9	28.5	28.3	28.0	27.8	27.8	27.8	(6)	
(7)		DBStd	1.11	1.00	0.92	1.11	1.04	1.00	0.98	1.15	1.04	0.86	0.84	0.99	0.98	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6711	561	525	593	576	587	556	567	568	538	552	535	553	(10)
(11)		CDD18.3	3669	303	292	334	326	328	306	309	309	288	293	285	294	(11)
(12)		CDH23.3	42473	3236	3334	4054	4058	4159	3765	3688	3721	3310	3141	2939	3067	(12)
(13)		CDH26.7	16044	1112	1336	1768	1793	1776	1509	1418	1400	1106	971	893	963	(13)
(14)	Wind	WSAvg	2.3	1.7	1.9	1.6	1.5	2.6	3.0	3.4	3.9	3.2	2.3	1.1	1.7	(14)
(15)	Precipitation	PrecAvg	2190	30	27	89	126	216	255	279	320	311	293	165	81	(15)
(16)		PrecMax	2676	156	84	325	272	471	429	473	556	460	509	308	226	(16)
(17)		PrecMin	1741	0	0	0	21	85	120	127	57	160	90	65	8	(17)
(18)		PrecStd	251	38	28	91	74	85	81	80	125	83	99	69	53	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.2	34.2	34.9	34.7	34.0	32.9	32.5	32.1	32.1	32.2	32.8	32.9	(19)
(20)			MCWB	25.7	25.1	26.0	27.4	28.1	28.1	27.6	27.5	27.6	27.0	27.0	25.8	(20)
(21)		2%	DB	32.5	33.5	33.9	33.7	33.0	32.1	31.7	31.5	31.2	31.4	31.9	32.1	(21)
(22)			MCWB	25.6	25.2	26.3	27.4	27.7	27.6	27.2	27.2	27.1	26.9	26.7	25.9	(22)
(23)		5%	DB	31.9	32.8	33.2	33.0	32.3	31.5	31.1	30.9	30.7	30.7	31.3	31.4	(23)
(24)			MCWB	25.4	25.4	26.5	27.3	27.6	27.4	27.0	26.9	26.9	26.6	26.5	25.8	(24)
(25)		10%	DB	31.1	32.0	32.4	32.4	31.6	31.0	30.6	30.5	30.2	30.1	30.6	30.7	(25)
(26)			MCWB	25.4	25.4	26.5	27.1	27.4	27.1	26.8	26.7	26.7	26.4	26.3	25.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.7	28.0	28.7	29.2	29.3	29.2	28.7	28.6	28.5	28.2	28.2	27.9	(27)
(28)			MCD B	30.8	32.0	32.3	32.5	32.0	31.6	31.2	31.0	31.2	30.6	31.4	31.1	(28)
(29)		2%	WB	27.6	27.2	28.1	28.6	28.9	28.5	28.1	28.0	27.9	27.7	27.5	27.1	(29)
(30)			MCD B	30.7	31.4	32.0	32.2	31.7	31.1	30.8	30.3	30.3	30.2	30.8	30.5	(30)
(31)		5%	WB	26.8	26.6	27.6	28.2	28.4	28.0	27.7	27.6	27.5	27.2	27.1	26.6	(31)
(32)			MCD B	30.0	30.9	31.6	31.8	31.2	30.8	30.4	30.0	29.9	29.8	30.3	29.9	(32)
(33)		10%	WB	26.1	26.0	27.0	27.6	27.9	27.6	27.2	27.3	27.1	26.9	26.6	26.1	(33)
(34)			MCD B	29.4	30.5	31.2	31.3	30.6	30.4	30.0	29.8	29.5	29.4	29.7	29.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.4	7.2	7.0	6.4	5.0	4.7	4.5	4.0	4.4	4.8	5.4	5.8	(35)
(36)			MCDBR	7.2	7.8	7.8	7.2	6.2	5.6	5.1	4.7	5.4	5.9	6.5	6.8	(36)
(37)			MCWBR	2.6	2.6	2.9	2.7	2.6	2.6	2.4	2.2	2.6	2.6	2.6	2.6	(37)
(38)		5% WB	MCDBR	6.6	7.2	7.2	6.7	5.7	5.4	4.8	4.3	5.0	5.4	6.2	6.2	(38)
(39)			MCWBR	3.3	3.3	3.3	3.1	3.2	2.9	2.7	2.4	2.8	2.8	2.8	3.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.461	0.475	0.486	0.474	0.451	0.456	0.459	0.453	0.441	0.444	0.450	0.457	(40)	
(41)		taud	2.314	2.245	2.253	2.293	2.391	2.399	2.377	2.383	2.411	2.404	2.372	2.341	(41)	
(42)		Ebn at Noon	856	853	845	843	844	829	830	849	871	871	859	851	(42)	
(43)		Edh at Noon	131	144	144	137	121	118	121	123	121	121	123	126	(43)	
(44)	All-Sky Solar Radiation	RadAvg	5.25	6.14	6.17	5.85	5.12	4.94	4.89	5.00	4.85	4.73	4.53	4.53	(44)	
(45)		RadStd	0.38	0.35	0.58	0.36	0.33	0.33	0.27	0.23	0.17	0.36	0.31	0.42	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)		+0.19	+0.25	N/A	N/A	+0.19	+0.21	N/A	N/A	+74	+75

Nomenclature: See separate page